

Grapes

Hollis Vane

ONE

Concord

Let me tell you about the grape.

I have spent most of my life around wine, and I have been paid, for a good part of it, to tell one wine from another. But the grape is not the wine. The grape is wine before it went to school. I want to tell you about the grape itself, the fruit, the berry, the thing on the vine before anyone has crushed it or bottled it or argued about it at dinner.

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp.¹ By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not.² The grape qualifies without argument.

The grape belongs to the genus *Vitis*, in the family Vitaceae, a

¹The ovary has two chambers, and each chamber holds two ovules, so that a grape can have at most four seeds. Seeded varieties have one to four. I have looked. I have not found a fifth.

²The strawberry is an accessory fruit. Most of its flesh grows from the receptacle, the part of the flower that holds the ovaries, and the specks on the outside are the true fruits, each one an achene.³ The raspberry is an aggregate: a cluster of many small fruits, each from a separate ovary of one flower.

³An achene is a small dry fruit with one seed that does not split open when ripe.

family of climbing plants.³ There are sixty to eighty species of *Vitis*, depending on whom you ask.⁴ Most of them grow in eastern Asia and in North America. Europe and western Asia have one. That one is *Vitis vinifera*, the wine-bearing vine, and it is responsible for the great majority of all the grapes grown on the planet.

The wild vine, from which the cultivated one was taken some eight thousand years ago in western Asia, comes in two sexes.⁵ The cultivated vine is hermaphrodite. Its flowers carry everything they need. It does not require a partner to set fruit. This is convenient for agriculture. It is convenient for the vine as well, though I doubt the vine has an opinion about it.

Ten thousand varieties of grape are known, according to the International Organisation of Vine and Wine. Thirteen of them cover more than a third of the world's vineyards. Thirty-three cover half.⁶ I will

³The family is large and mostly unfamiliar. The *Encyclopædia Britannica* counts twelve genera in it; a group of botanists writing in 2018 (Wen and others, in the *Journal of Systematics and Evolution*) count sixteen genera and about nine hundred and fifty species. Virginia creeper is a member. So is Boston ivy, which is not from Boston and is not an ivy.

⁴The *Britannica* gives the range; the International Organisation of Vine and Wine says eighty, on the authority of a 2007 paper by Wen and others. The disagreement is about the Asian species, which are many and poorly separated.

⁵The wild form is *Vitis vinifera* subspecies *sylvestris*; the cultivated form is subspecies *vinifera*. The usual account, as of 2020, places the domestication in the Near East (Myles and others, *Proceedings of the National Academy of Sciences*, 2011). On the sexes: Vasconcelos and others, *American Journal of Enology and Viticulture* (2009), and Picq and others, *BMC Plant Biology* (2014).^a

⁶The Organisation's 2017 report, *Distribution of the World's Grapevine Varieties*. The same report adds that the standard catalogue, the *Vitis International Variety Catalogue*, held 21,045 names, of which 12,250 were for *vinifera*, but many of these are the same grape under different names, and the true number of *vinifera* varieties is nearer six thousand.^b

^aPicq and his colleagues traced the difference to a small region of a single chromosome.

^bThe same vine may be called one thing in its own valley, another two valleys over, and a third in a nurseryman's catalogue in another language. A grape that travels

repeat the numbers because they deserve it. Ten thousand varieties. Thirty-three of them on half the vineyard land on earth.

How much grape is there in the world? In 2018, the most recent year for which I have figures, the world grew 77.8 million metric tonnes.⁷

The skin of a grape carries a bloom, a dull waxy coating that rubs off on the thumb. It is there to keep the water in. I once spent an afternoon rubbing the bloom off Cabernet berries one by one with a damp cloth, to see what the grape looked like underneath. It looked like a grape, only shinier and slightly sticky.

I bought a bunch of grapes yesterday at the market in St. Helena. It is January, and nothing in this valley is ripe; the vines outside my window are bare wood. The box said *Product of Chile*, which means it was summer where these grapes were picked, and they crossed the equator to be eaten in the winter of someone who would never ask their name.

I weighed them: 612 grams. I counted the berries. There were 117. I counted again and got 118. A third time, slowly, touching each berry with the tip of a pencil: 117.

☛ Three counts and two answers. I am a careful man. I have been

⁷International Organisation of Vine and Wine, *2019 Statistical Report on World Vitiviniculture*. Of the 77.8 million tonnes, 57 per cent went to wine (and to musts and juices), 36 per cent were eaten as table grapes, and 7 per cent were dried — counted as fresh grapes, at about four kilograms to one kilogram of raisins. The largest producers, in millions of tonnes: China 11.7, Italy 8.6, the United States 6.9, Spain 6.9, France 6.2. The vineyards covered 7.4 million hectares.^a

collects names, as any traveller does. The catalogue's work is to find out which names belong to one plant.

^aIn pounds — which is how I was raised to think, and how I still think when I am tired — 77.8 million tonnes is about 171.5 billion pounds. Divide that among the 7.7 billion people the United Nations counted in 2019 and it comes to about 22 pounds a head. I have done the division three times and it comes out the same each time, which is more than I can say for all of my counting.

paid, for most of my life, to be careful. And I could not count a bunch of grapes twice and get the same number. I will believe 117, because I got it twice.

A bunch of grapes is, botanically, a panicle: a branched stalk carrying many flowers, and later many fruits.⁸

Then, because I was not tired, I cut eleven of the berries in half with a paring knife on a white plate. These were sold as seedless, and the box was right: eleven berries, zero seeds, eleven times.

The first grape I ever knew was not a grape. It was a jelly.

I grew up in Iowa, in a town I will not bother to name. Name it and someone from there will write to correct me about the year the elevator burned; leave it unnamed and it could be any of a thousand towns, which is nearer the truth. There was a grain elevator, a main street, a church, a café, and a hardware store. In August the heat came up out of the fields and stood in the street. There were no vines. If you had asked me, at ten, what a grape was, I would have pointed at a jar.

The jar said *Welch's*. The jelly inside it was purple, nearly black in the jar and a bright translucent violet when you spread it thin. My mother did not spread it thin.⁹ On white bread, with peanut butter, it was the only grape I knew.¹⁰

⁸The stalk that joins the bunch to the cane is the peduncle; the main axis is the rachis; each berry hangs from a short stem of its own, the pedicel. (The terms are in Winkler's *General Viticulture*, the standard textbook.) A single cluster may carry as many as a thousand flowers. If one in three sets fruit, the growers are content.^a

⁹In the early 1960s Welch's sold its grape jelly in drinking glasses printed with the Flintstones. When the jelly was gone the glass was yours. I was too old to say I wanted one, so I drank from Fred Flintstone's face only when nobody was in the kitchen.

¹⁰The peanut-butter-and-jelly sandwich, which Americans sometimes credit to soldiers in the Second World War, was not invented by them. The first published recipe appeared in 1901, in the *Boston Cooking-School Magazine*, by Julia Davis Chan-

^aBy that arithmetic my 117 berries were once about 350 flowers, and some 230 of those came to nothing.

That jelly is made from one grape: the Concord. And the taste of it — sweet, heavy, a little wild — is the taste that Americans of my generation mean when they say “grape.” It is not the taste of the grape of wine. A European will tell you, politely, that it tastes of something else. The word for the something else is *foxy*. The French, who borrowed it from the English “fox grape,” say *foxé*. The molecule chiefly responsible has a name: methyl anthranilate.¹¹ One molecule is chiefly responsible for the difference between the grape of a child’s lunch box and the grape of a Burgundy vineyard.

The Concord grape has a maker. His name was Ephraim Wales Bull, and he was born in Boston in 1806.

He moved to Concord, Massachusetts, in 1836, to a house on Lexington Road. By his own account he planted seeds from a wild grape in the autumn of 1843, chose among the seedlings, and waited. His vine first bore fruit in 1849. He showed the grape to the Massachusetts Horticultural Society in 1853, and in 1854 C. M. Hovey of Boston put it on the market.¹²

What the country wanted, Bull said, was “a good table and wine grape, which shall also be early, hardy and prolific.” The European vines that people had tried to grow in New England died in the winter

dler, and it called for currant or crab-apple jelly, not grape. The sandwich became a staple of American childhood during the Depression, when bread came sliced and peanut butter came cheap. The soldiers ate it; they did not invent it.

¹¹ Methyl anthranilate was first found in grape juice in 1921, by Power and Chesnut, in the *Journal of the American Chemical Society*; they found it in the American grapes and not in the European. A later paper (Robinson and others, *American Journal of Enology and Viticulture*, 2014) still calls it the compound “considered to be responsible for the distinctive ‘foxy’ aroma and flavor” of the Concord. Other compounds help; this one leads.

¹² The dates are in the Concord Free Public Library’s papers on Bull and in the histories kept by the USDA’s National Agricultural Library. The 1843 sowing is Bull’s own account, quoted at second hand; he also spoke of raising two generations of seedlings, which does not sit easily in six years. Where the sources disagree I have given the date they agree on.

or would not ripen before the frost. The Concord ripened early and survived. Americans planted it widely.

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Bull served in the state legislature. He did not grow rich. He died in 1895, in the Concord Home for the Aged, and he is buried in Sleepy Hollow Cemetery under a stone that reads *He sowed, others reaped*.¹⁴

Others reaped. The chief of them was a dentist. Thomas Bramwell Welch was born in England, a Methodist and a man who believed that alcohol was the ruin of the neighbourhood. He practised dentistry in Vineland, New Jersey. In 1869 he heated the juice of Concord grapes to keep it from fermenting, so that his church could take communion without taking wine.¹⁵

Bull grew the grape. Welch squeezed it. We remember the squeezer. The grower lies under a stone in Concord with an epitaph that nobody can punctuate. I do not say this to blame Welch; he did what there was to do, and he did it for his church.

The bunch is still on my kitchen table, minus eleven berries, in its plastic box from Chile. In the morning I will count what is left. I expect 106: the 117 I believe in, minus the eleven I cut. If I get 106, I will believe it at once. If I do not, I will count again until I get a number I can accept, and then I will stop.

That is how this book will go.

¹³The Concord is usually called a cultivar of *Vitis labrusca*, the American fox grape. But its parentage includes some *vinifera*: Huber and others (*Vitis*, 2016) make it an offspring of ‘Catawba,’ which has European blood, and an unknown wild parent.^a

¹⁴The punctuation of the epitaph varies from source to source: a full stop, a comma, nothing. I have not seen the stone. I mean to.

¹⁵Welch’s says he used forty pounds of grapes from the vines in front of his house. In 1897 the business built a large juice plant in Westfield, New York, in the strip of Concord vineyards along the shore of Lake Erie still called the Grape Belt. Its grape spread, Grapelade, dates from 1918; its grape jelly, from 1923. The jelly is still made from Concord grapes. It says so on the jar.

^aSo the most American of grapes is not entirely American.